

Expanding the chemistry of single-ion conducting poly(ionic liquid)s with polyhedral boron anions

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Running title

Lithium-ion conducting poly(ionic liquid)s with boron anions

Abstract

In this work the synthesis of a new family of single-ion conducting random copolymers bearing polyhedral boron anions is reported. For this purpose two novel ionic monomers, namely $[B_{12}H_{11}(OCH_2CH_2)_2-OC(=O)-C(CH_3)=CH_2]^2-[(C_4H_9)_4N^+]_2$ and $[8-(OCH_2CH_2)_2-OC(=O)-C(CH_3)=CH_2-3,3'-Co(1,2-C_2B_9H_{10})(1',2'-C_2B_9H_{11})]^-K^+$, having methacrylate function, diethylene glycol bridge and *closo*-dodecaborate or cobalt bis(1,2-dicarbollide) anions were designed. Such monomers differ from previously reported ones by (1) chemically attached highly delocalized boron anions, by (2) valency of the anion (divalent-anion and monovalent one) and by (3) the presence of oxyethylene flexible spacer between the methacrylate group and bonded anion. Their free radical copolymerization with poly(ethylene glycol) methyl ether methacrylate and subsequent ion exchange provided lithium-ion conducting polyelectrolytes showing low T_g ($-53 \div -49^\circ C$), ionic conductivity up to $9.1 \times 10^{-7} S\ cm^{-1}$, lithium transference number up to 0.61 ($70^\circ C$) and electrochemical stability up to 4.1 V vs. Li^+/Li ($70^\circ C$). The incorporation of propylene carbonate (20 - 40 wt%) into the copolymers resulted in the enhancement of their ionic conductivity by one order of magnitude and significantly increased their electrochemical stability up to 4.7 V vs. Li^+/Li ($70^\circ C$).

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This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1002/pi.5878

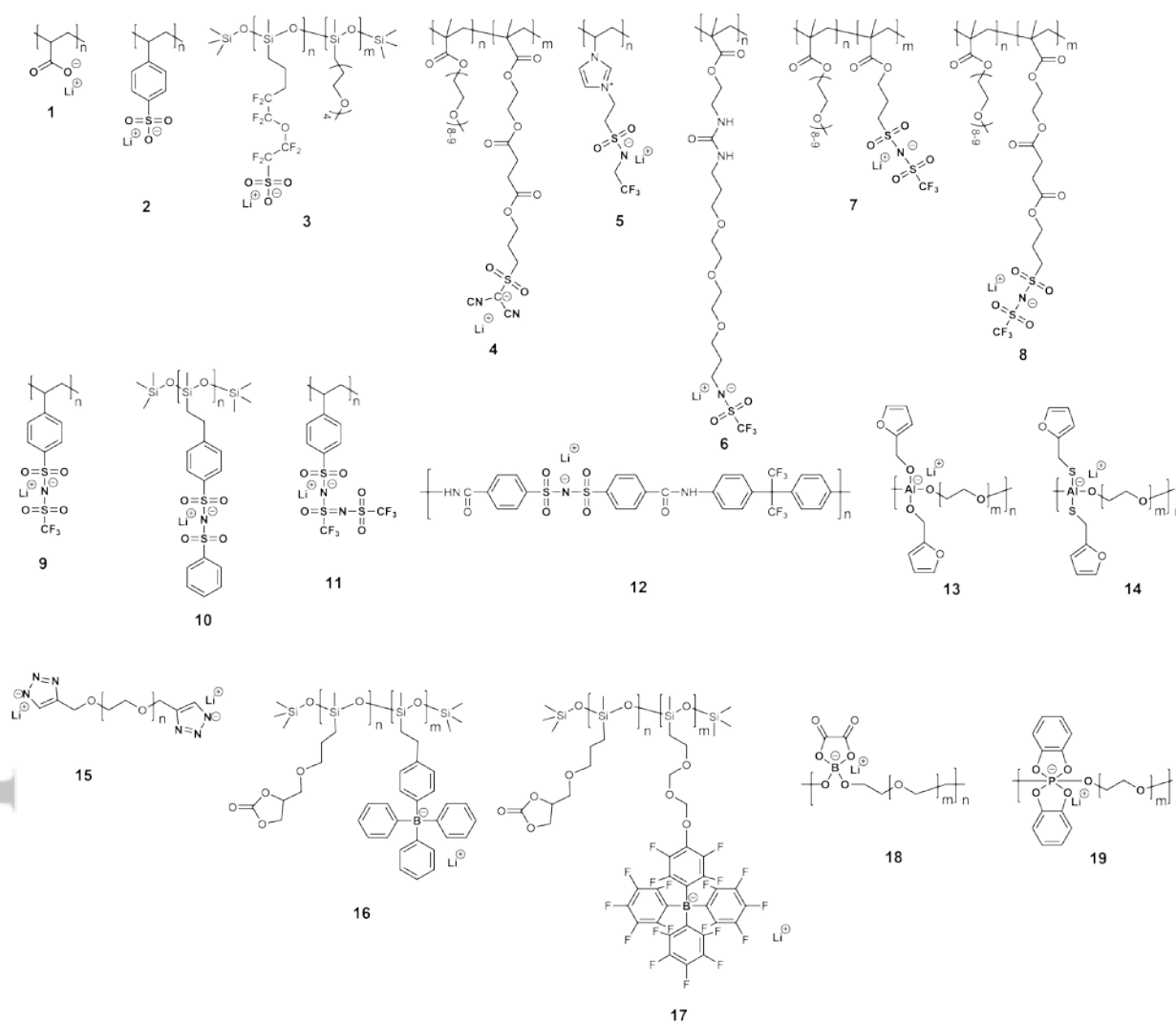
Keywords: poly(ionic liquid)s, single-ion conductors, polyelectrolytes, *closo*-dodecaborate anion, cobalt bis(1,2-dicarbollide) anion, Li batteries

INTRODUCTION

Currently lithium-ion batteries power many of today's electric vehicles and portable electronics. They offer high energy density and high cell voltage of 3.6 volts that allows battery pack to be design with only one cell in a lightweight package¹⁻⁴. Low maintenance and the absence of memory effect are among other important advantages proposed by Li batteries. In addition, the self-discharge in Li cells is much less in comparison with other types of commercial batteries, making them well suited for modern fuel gauge applications. However, the existent technology is based mostly on volatile and flammable liquid electrolytes posing potentially serious fire hazard in case of cell overheating. For instance, accidental overheating of the battery may cause decomposition reactions of the electrolyte followed by an uncontrolled temperature rise and battery failure, eventually, flames or explosions. This mechanism is usually referred as "thermal runaway"⁵. Thus, the development of solid polymer electrolytes (SPEs) as a safer alternative to conventional electrolytes deserves additional attention^{6,7}.

Among candidates for SPEs, a new class of polyelectrolytes, namely poly(ionic liquid)s (PILs), is recently intensively investigated⁸⁻¹⁵. PILs are macromolecular analogues of ionic liquids (ILs) that ideally merge the best attributes of ILs (high thermal & (electro-)chemical stability, enhanced ionic conductivity, etc.) and polymers (processability, viscoelasticity, film-forming properties, etc.)^{8,9,13}. Among the variety of PILs those polyelectrolytes, the so-called "single-ion conductors (SICs)", carrying anionic functional groups chemically bonded to the polymer backbone and bearing lithium ions as the mobile counterpart, are of particular interest for batteries applications¹⁶. Such single lithium-ion conducting anionic PILs (Scheme 1) offer several advantages: 1) the increase of Li transference number to unity¹⁷⁻¹⁹; 2) the minimization of the dendrite growth^{16,20}; 3) the increase of the oxidative stability^{16,21,22} and 4) the support of high charge/discharge rates²³. The described excellence is mainly governed by the fixation (chemical bonding) of the anion to the polymer backbone and subsequently the limitation of its migration to the anode's surface where it can electrochemically degrade²⁴.

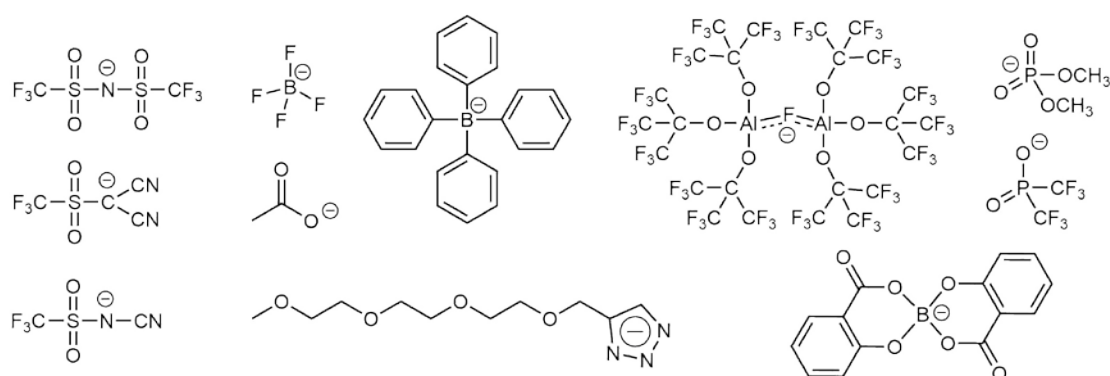
It was revealed that the ionic conductivity of SICs is generally determined by the nature of the attached anion, the position of the anion (backbone versus side chain), the spacer length, the nature of the polymer backbone and the macromolecular architecture (random or block copolymers, branched or linear polyelectrolyte, etc.)^{16,8}. The investigation of single ion conductors started from polymers bearing strongly coordinating anions, such as carboxylates and sulfonates (Scheme 1, **1**^{25,26} and **2**²⁷). As the gained mobility of lithium cations was not high enough, the attempt to increase the conductivity by transfer from sulfonate to its perfluorinated analog has been performed (Scheme 1, **3**²⁸).



Scheme 1. Typical examples illustrating the structural variety of anions in single-ion conducting PILs.

Next studies were further focused on charge delocalization by bonding of the electron withdrawing groups to sulfonate fragment (Scheme 1, **4**¹⁹, **5**²⁹, **6**³⁰). Afterwards the era of SICs

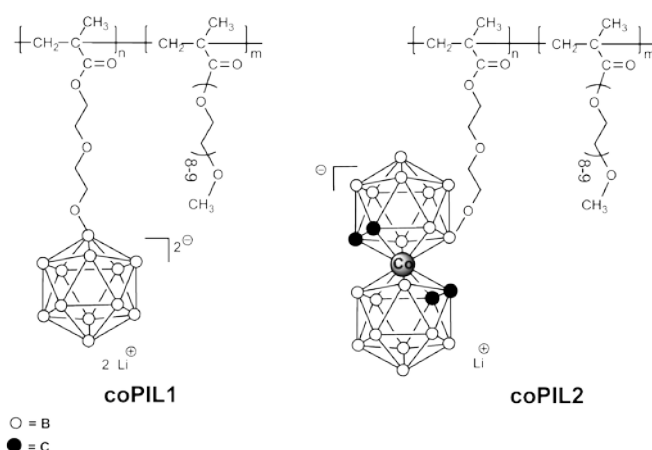
incorporating anions structurally similar to famous bis(trifluoromethylsulfonyl)imide (TFSI) ion has started (Scheme 1, **7**¹⁷, **8**¹⁹, **9**^{23,31,32}, **10**³³, **12**³⁴). It was later extended by the development of PIL bearing the so-called “super-TFSI” anion (Scheme 1, **11**³⁵). The TFSI- and super-TFSI-derivatives have been shown to possess higher ion dissociation, resulting in higher ionic conductivities than the PILs with unmodified sulfonate anions. However, only a few examples of SICs having anions other than the sulfonate derivatives have been studied (Scheme 1, **13-19**). Among them are the polyelectrolytes with complex aluminate ([Al(OR)₄]⁻) and thioaluminate ([Al(SR)₄]⁻) anions (Scheme 1, **13**³⁶, **14**³⁶), 1,2,3-triazolate anions (Scheme 1, **15**³⁷) various borate (Scheme 1, **16**³⁸, **17**³⁹, **18**²⁰) and phosphate (Scheme 1, **19**⁴⁰) anions. These PILs can be represented as the macromolecular analogs of ionic liquids with [((CF₃)₃CO)₃Al-F-Al{OC(CF₃)₃}₃]⁻⁴¹, triazolate⁴², BF₄⁻⁴³, BPh₄⁻⁴⁴, bis(salicyl)borate⁴⁵, PO₂(CF₃)₂⁻⁴⁶ and PO₄(CH₃)₂⁻⁴⁷ anions, correspondingly (Scheme 2).



Scheme 2. Examples of anions used in ILs synthesis and served as an inspiration for PILs design.

Furthermore, it was revealed that to increase the ion conductivity it is preferably to place the anion in the side chain rather than in polymer backbone (Scheme 1, **7**¹⁷, **9**^{23,31,32}, **11**³⁵ and **12**³⁴). The longer is the spacer between the attached anion and main polymer chain the higher is the conductivity (Scheme 1, **7**¹⁷ and **8**¹⁹). The transfer from random copolymers to complex macromolecular architectures (block or triblock copolymers) using reversible addition-fragmentation chain-transfer (RAFT)^{17,18,48} or nitroxide-mediated polymerization (NMP)^{23,32,49,50} allowed to increase ion mobility and to achieve as high ionic conductivity in a solid state as 3×10⁻⁶ S cm⁻¹.

In spite of a great success gained with SICs, the goal to achieve the conductivity level of $10^{-3} \div 10^{-4} \text{ S cm}^{-1}$ (at 25°C) suitable for industrial applications still remains unreachable¹⁶. One of the possible solutions for enhancement of Li mobility in solid polyelectrolytes can be the development of novel bulky weakly- or even non-coordinating anions. Thus, by analogy with low melting carborane based ionic liquids^{51,52}, in this work we present the strategy to introduce weakly-coordinating polyhedral boron anions⁵³, namely *closo*-dodecaborate and cobalt bis(1,2-dicarbollide) ones, into the structure of PILs (Scheme 3). Two novel ionic-liquid-like methacrylic monomers (ILMs) were synthesized and further copolymerized with poly(ethylene glycol) methyl ether methacrylate (PEGM) via free radical polymerization. This afforded a series of high molecular weight ($M_w = 0.94 - 4.57 \times 10^{-5} \text{ g mol}^{-1}$) random copolymers showing low T_g ($-53 \div -49^\circ\text{C}$), ionic conductivity up to $9.7 \times 10^{-7} \text{ S cm}^{-1}$ at 25 °C and electrochemical stability in the range of $3.9 \div 4.1 \text{ V}$ vs. Li^+/Li .



Scheme 3. Chemical structures of the novel anionic copolymers synthesized in the present study ($n:m = 1:7.2$ by mol).

MATERIALS AND METHODS

Materials

Poly(ethylene glycol) methyl ether methacrylate (PEGM, $M_w = 475 \text{ g mol}^{-1}$, Aldrich), 4-methoxyphenol (99%, Acros), tetrabutylammonium bromide (99+%, Acros), lithium chloride (99%, Acros), acetonitrile (ACN, HPLC grade 99%, Acros), dimethyl formamide (DMF, 99.8%, anhydrous, Acros), lithium metal foil (Chemetall Foote Corporation) were used without further purification. The Spectra/Por 1 (Spectrumlabs) dialysis tubing with MWCO 6000-8000 Dalton were used for polymer dialysis.

1,4-Dioxane was distilled from deep purple sodium-benzophenone complex. Methacryloyl chloride (>97%, Aldrich) was distilled before use. Triethylamine (98+%, Merck) and diethyl ether were distilled over NaOH and P₂O₅, respectively. Potassium carbonate (99+%, anhydrous, Acros) was dried at 140°C and after cooling to room temperature was immediately used in reaction. 2,2'-Azobisisobutyronitrile (AIBN, initiator, 98%, Acros) was recrystallized from methanol.

[B₁₂H₁₁O(CH₂CH₂)₂O]⁺[(C₄H₉)₄N]⁺ (**20**)⁵⁴ and [8-O(CH₂CH₂)₂O-3,3'-Co(1,2-C₂B₉H₁₀)(1',2'-C₂B₉H₁₁)] (**23**)⁵⁵ were synthesized in accordance with the previously published procedures.

Synthesis of [B₁₂H₁₁(OCH₂CH₂)₂-OH]²⁻[(C₄H₉)₄N⁺]₂ (**21**)

[B₁₂H₁₁O(CH₂CH₂)₂O]⁺[(C₄H₉)₄N]⁺ (**20**) (8.89 g, 0.019 mol) was added to the 500 mL of aqueous solution of NaOH (4.53 g, 0.113 mol) under stirring at room temperature. The suspension was then refluxed for 32 h with gradual addition of 250 mL of distilled H₂O until the solution was formed. After cooling down to RT, it was filtered from insoluble residue and neutralized with 12.4N aq. hydrochloric acid to pH = 7. To the obtained neutral solution, the aqueous solution of (C₄H₉)₄NBr (15.20 g, 0.047 mol) in 30 mL of H₂O was added dropwise at room temperature and stirred for 2 h. Then the solution was concentrated at 60°C under reduced pressure to ~250 mL, cooled down to RT and put in the fridge for 12 h. The precipitated white crystals were collected by filtration, washed with cold water (4×15mL) and dried at 50°C/10 mm Hg for 6 h (hereinafter with a special flask filled with P₂O₅ and introduced into the vacuum line). Further on crystals were redissolved in 350 mL of CH₂Cl₂ and washed with distilled water (4×100 mL). The CH₂Cl₂ solution was dried over anhydrous MgSO₄, the magnesium sulfate was filtered off and dichloromethane was stripped off under reduced pressure. Product was obtained as white powder, which was dried at 50°C/10 mm Hg for 3 h and finally at 50°C/1 mm Hg for 10 h. Yield: 9.26 g (67%); melting point (mp) = 133-135°C; ¹H NMR, 400.1 MHz, DMSO-d₆, δ ppm: 4.61 (s, 1H, CH₂-OH), 3.45 (m, 2H, B₁₂H₁₁-O-CH₂), 3.36 (m, 6H, -O-CH₂CH₂-O-CH₂CH₂-), 3.17 (m, 16H, Bu₄N⁺), 1.57 (m, 16H, Bu₄N⁺), 1.33 (m, 16H, Bu₄N⁺), 0.94 (t, 24H, Bu₄N⁺); ¹¹B NMR, 128.4 MHz, DMSO-d₆, δ ppm: 6.5 (s, 1B), -16.8 (d, 5B), -18.3 (d, 5B), -23.1 (d, 1B); IR (KBr), cm⁻¹: 3309 (w, νOH), 2961 (vs, νCH), 2873 (s, νCH), 2474 (vs, νBH), 1473 (m), 1383 (w), 1158 (m, νC-O-C), 1115 (m, νC-OH), 1050 (m, νC-O-C), 1021 (m, νC-O-C), 886 (w), 733 (w); Calc. for C₃₆H₉₂B₁₂N₂O₃ (730.9): C, 59.16%; H, 12.69%; N, 3.83%; Found: C, 59.11%; H, 12.64%; N, 3.77%.

Synthesis of [B₁₂H₁₁(OCH₂CH₂)₂-OC(=O)-C(CH₃)=CH₂]²⁻[(C₄H₉)₄N⁺]₂ (**22**)

Freshly distilled triethylamine (1.53 g, 0.015 mol) and **21** (9.20 g, 0.013 mol) were dissolved in 30 mL of anhydrous acetonitrile with stirring under inert atmosphere and the solution was cooled in

an ice bath to 0-3°C. The solution of methacryloyl chloride (1.58 g, 0.015 mol) in 8 mL of anhydrous acetonitrile was slowly added dropwise to the cooled solution of **21** and TEA with gentle stirring. The stirring was then continued at 0-1°C for 1 h, whereupon the ice bath was removed and the reaction mixture was allowed to warm up to RT. The reaction proceeded at 20°C for 2 h, whereupon the precipitate was filtered off and the solution was concentrated at 30°C under reduced pressure to ~15 mL. It was further precipitated into 100 mL of anhydrous diethyl ether and the obtained viscous white oil was washed with Et₂O (4×100 mL) and afterwards dried at 25°C/10 mm Hg for 3 h. The product was redissolved in 250 mL of CH₂Cl₂ and washed with distilled water (5×100 mL). The CH₂Cl₂ solution was dried over anhydrous MgSO₄, the magnesium sulfate was filtered off, the catalytic amount of 4-methoxyphenol (as an inhibitor) was added and dichloromethane was stripped off under reduced pressure. Product was obtained as colorless viscous oil, which was dried at 50°C/1 mm Hg for 5 h. After 3 months storage at 0-5°C the viscous oil crystallized into the white wax like crystals. Yield: 9.83 g (98%); mp = 70-74°C; ¹H NMR, 400.1 MHz, acetone-d₆, δ ppm: 6.05 (s, 1H, CH₂=C(CH₃)–), 5.59 (s, 1H, CH₂=C(CH₃)–), 4.20 (t, 2H, –O–CH₂CH₂–OC(O)–, J = 5 Hz), 3.70 (t, 2H, –O–CH₂CH₂–O–CH₂CH₂–OC(O)–, J = 5 Hz), 3.64 (m, 4H, B₁₂H₁₁–O–CH₂CH₂–O–), 3.52 (t, 4H, B₁₂H₁₁–O–CH₂CH₂–O–, J = 6 Hz), 3.40 (m, 16H, Bu₄N⁺), 1.88 (s, 3H, CH₂=C(CH₃)–), 1.75 (m, 16H, Bu₄N⁺), 1.41 (m, 16H, Bu₄N⁺), 0.96 (t, 24H, Bu₄N⁺); ¹¹B NMR, 128.4 MHz, acetone-d₆, δ ppm: 6.6 (s, 1B), –16.5 (d, 5B, J = 158 Hz), –17.9 (d, 5B, J = 149 Hz), –22.7 (d, 1B, J = 128 Hz); IR (KBr), cm^{–1}: 2963 (vs, νCH), 2874 (s, νCH), 2472 (vs, νBH), 1714 (m, νC=O), 1630 (w, νC=C), 1476 (m), 1381 (w), 1168 (m, νC–O–C), 1128 (w), 1048 (m, νC–O–C), 1021 (m, νC–O–C), 882 (w), 732 (w); Calc. for C₄₀H₉₆B₁₂N₂O₄ (798.9): C, 60.14%; H, 12.11%; N, 3.51%; Found: C, 59.63%; H, 12.71%; N, 3.41%.

Synthesis of [8-(OCH₂CH₂)₂–OC(=O)–C(CH₃)=CH₂–3,3'–Co(1,2–C₂B₉H₁₀)(1',2'–C₂B₉H₁₁)]⁺K⁺ (**24**)

Compound **23** (1.00 g, 2.44 mmol) was dissolved in 20 mL of anhydrous acetonitrile and the freshly dried K₂CO₃ (1.70 g, 12.18 mmol) was added in one portion. To the obtained suspension a solution of methacrylic acid (0.21 g, 2.44 mmol) in 30 mL of acetonitrile was added dropwise and stirring was continued at RT for 60 h under inert atmosphere. Afterwards the precipitate was filtered off, acetonitrile was stripped off under reduced pressure without heating and the resultant orange oily residue was purified by column chromatography using dichloromethane/hexane (3:2 by volume) as the mobile phase. The catalytic amount of 4-methoxyphenol (as an inhibitor) was added to the solution and solvents were stripped off under reduced pressure. Product in a form of light orange powder dried at 40°C/1 mm Hg for 12 h. Yield: 1.28 g (98 %); mp = 176-178°C; ¹H

NMR, 400.1 MHz, acetone-d₆, δ ppm: 6.10 (s, 1H, CH₂=C(CH₃)–), 5.61 (s, 1H, CH₂=C(CH₃)–), 4.25 (m, 6H, O-CH₂CH₂-O-(C=O) + CH_{carborane}), 3.74 (t, 2H, -CH₂CH₂-O-CH₂CH₂-O-(C=O), *J* = 4 Hz), 3.60 (t, 2H, -CH₂CH₂-O-CH₂CH₂-O-(C=O), *J* = 4 Hz), 3.56 (t, 2H, -CH₂CH₂-O-CH₂CH₂-O-(C=O), *J* = 4 Hz), 1.93 (s, 3H, CH₂=C(CH₃)–); ¹³C NMR, 100.0 MHz, acetone-d₆, δ ppm: 166.7 (-OC(=O)-), 136.5 (C=CH₂), 124.9 (C=C₂), 71.9 (OCH₂), 68.8 (OCH₂), 68.5 (OCH₂), 64.0 (OCH₂), 54.3 (C_{carb}), 46.4 (C_{carb}), 17.6 (CH₃); ¹¹B NMR, 128.4 MHz, acetone-d₆, δ ppm: 23.0 (s, 1B), 4.1 (d, 1B, *J* = 140 Hz), 0.4 (d, 1B, *J* = 141 Hz), -2.5 (d, 1B, *J* = 152 Hz), -4.3 (d, 2B, *J* = 152 Hz), -7.4 (d, 2B, *J* = 132 Hz), -8.3 (d, 4B, *J* = 125 Hz), -17.4 (d, 2B, *J* = 155 Hz), -20.5 (d, 2B, *J* = 161 Hz), -22.1 (d, 1B), -28.6 (d, 1B, *J* = 162 Hz); IR (KBr), cm⁻¹: 3032 (w, νCH_{carb}), 2902 (m, νCH), 2870 (s, νCH), 2564 (vs, νBH), 1716 (vs, νC=O), 1630 (m, νC=C), 1453 (m), 1315 (m), 1248 (w), 1158 (s, νC-O-C), 1117 (s), 1084 (vs, νC-O-C), 1044 (m, νC-O-C), 995 (m), 975 (m), 952 (m), 884 (w), 819 (m), 751 (w); Calcd. for C₁₂H₃₄B₁₈O₄CoK (535.0): C, 26.94%; H, 6.41%; B, 36.37%; Found: C, 27.04%; H, 6.25%; B, 35.30%.

Polymer synthesis

The example is given for the synthesis of **coPIL1** copolymer with the composition of [22]:[PEGM] = 1:7.2 by mol. Ionic monomer **22** (0.67 g, 0.835 mmol), PEGM (2.85 g, 6.01 mmol), DMF (7.00 g) and AIBN (0.0352 g, 1.0 wt.%) were gently mixed in a Schlenk flask at ambient temperature. After triple freeze-thaw-pump cycles the flask was filled with argon and heated at 60 °C for 6 h. The resultant transparent and highly viscous polymer solution was diluted with water, dialyzed against 0.5M aq. LiCl and further with water for 3 days, whereupon it was freeze-dried. Resultant polymer represented a soft sticky colorless material that was thoroughly dried at 50°C/1 mm Hg for 24 h and finally stored in the argon filled glove box for 10 days prior to further investigation. Yield: 2.00 g (57%); ⁷Li NMR, 155.5 MHz, DMSO-d₆, δ ppm: -0.92; ¹¹B NMR, 128.4 MHz, DMSO-d₆, δ ppm: 6.5 (s, 1B), -16.7 (m, 5B), -18.2 (m, 5B), -23.1 (m, 1B); IR (KBr), cm⁻¹: 2873 (vs, νCH), 2474 (vs, νBH), 1729 (vs, νC=O), 1455 (s), 1350 (s), 1249 (s), 1107 (vs, νC-O-C), 1042 (m, νC-O-C), 950 (s), 854 (m), 748 (w); Calcd. for (C₈H₂₄B₁₂Li₂O₄)₁×(C₂₃H₄₄O₁₀)_{8.5} (4412.9): C, 55.39%; H, 9.09%; B, 2.94%; Found: C, 54.97%; H, 9.16%; B, 2.68%; *M_w* = 3.16×10⁵, *M_w*/*M_n* = 3.54 (GPC); *T_g* = -53°C (DSC); σ_{DC} = 1.5×10⁻⁷ S cm⁻¹ (25 °C).

For **coPIL2** the same [24]:[PEGM] = 1:7.2 by mol. ratio was applied. The obtained polymer represented a soft sticky bright orange material. Yield: 3.53 g (76%); ⁷Li NMR, 155.5 MHz, D₂O, δ ppm: 0.12; ¹¹B NMR, 192.5 MHz, D₂O, δ ppm: 23.7, 4.7, -2.5, -4.9, -7.8, -18.3, -21.2, -30.3; IR (KBr), cm⁻¹: 2874 (vs, νCH), 2558 (vs, νBH), 1729 (vs, νC=O), 1453 (s), 1350 (m), 1249 (s), 1107

(vs, ν C-O-C), 1038 (m, ν C-O-C), 949 (m), 853 (m), 748 (w); Calcd. for $(\text{C}_{12}\text{H}_{34}\text{B}_{18}\text{CoLiO}_4)_1 \times (\text{C}_{23}\text{H}_{44}\text{O}_{10})_{6.75}$ (3746.9): C, 53.61%; H, 8.90%; B, 5.19%; Found: C, 53.82%; H, 9.05%; B, 5.63%; $M_w = 2.55 \times 10^5$, $M_w/M_n = 4.19$ (GPC); $T_g = -49^\circ\text{C}$ (DSC); $\sigma_{\text{DC}} = 9.1 \times 10^{-7} \text{ S cm}^{-1}$ (25 $^\circ\text{C}$).

Characterization

NMR spectra were recorded on AMX-400 and Avance II 500 MHz spectrometers (Bruker) at 25 $^\circ\text{C}$ in the indicated deuterated solvents and are listed in ppm. The signal corresponding to the residual protons of the deuterated solvent was used as an internal standard for ^1H and ^{13}C NMR, while the CHCl_2F , LiNO_3 and $\text{F}_3\text{B} \cdot \text{OEt}_2$ were utilized as external standards for ^{19}F , ^7Li and ^{11}B NMR, respectively. IR spectra were acquired on a Nicolet Magna-750 Fourier IR-spectrometer using KBr pellets (128 scans, resolution is 2 cm^{-1}).

Inherent viscosities (η_{inh}) of copolymers were measured using an Ostwald capillary viscometer at 25.0 $^\circ\text{C}$ for 0.05 g of polymer in 10.0 mL of aq. 0.5 M LiCl solution.

A 1200 Infinity gel permeation chromatograph (GPC, Agilent Technologies) was used to determine M_n , M_w and PDI of the copolymers. The chromatograph was equipped with an integrated IR detector, a PL PolarGel-M column and a PL PolarGel-M guard column (Agilent Technologies). The eluent was a 0.1 M LiCl solution in mixture of water/ACN (4:1 v/v) and the flow rate was of 1.0 mL min^{-1} at 25 $^\circ\text{C}$. PEG standards (Kit Poly(ethylene oxide) PSS company, $M_p = 26.1 - 969 \times 10^3$) were used to perform calibration.

Differential Scanning Calorimetry (DSC) experiments were performed on a DSC3+ STARe System differential calorimeter (Mettler Toledo) with a heating rate of 10 $^\circ\text{C min}^{-1}$ in the range of -80 to 80 $^\circ\text{C}$. Glass transition temperatures (T_g) were determined during second heating cycle. coPILs samples were hermetically sealed in aluminum pans inside argon-filled glove-box (MBRAUN MB-Labstar, H_2O and O_2 content <0.5 ppm). Thermal gravimetric analysis (TGA) was carried out in air on a TGA2 STARe System (Mettler Toledo) applying a heating rate of 5 $^\circ\text{C min}^{-1}$.

Ionic conductivity (σ) was determined by electrochemical impedance spectroscopy (EIS) with a VSP potentiostat/galvanostat (Bio-Logic Science Instruments). To avoid any influence of moisture/humidity on the conductivity of polymer electrolytes, the latter were preliminary dried at 60 $^\circ\text{C}/1 \text{ mm Hg}$ for 12 h in the B-585 oven (Buchi Glass Drying Oven, Switzerland) filled with P_2O_5 and were transferred under vacuum inside an argon-filled glovebox. Polymers were sandwiched between two stainless steel electrodes. The distance between the electrodes was kept equal to 250 μm using a Teflon spacer ring with the inner area of 0.502 cm^2 . Symmetrical stainless

steel/**coPIL**/stainless steel assembly was clamped into the 2032 coin cell and afterwards was taken out from glovebox. EIS experiments were carried by applying a 10 mV perturbation in the frequency range from 10^{-2} to 2×10^5 Hz and in a temperature range from 20 to 100 °C. Temperature was controlled using the programmed M-53 oven (Binder, Germany), where cells were allowed to reach thermal equilibrium for at least 30 minutes before each test.

Cyclic voltammetry (CV) was used to determine the electrochemical stability window of the solid polymer electrolytes at 70 °C. VSP potentiostat/galvanostat (Bio-Logic Science Instruments, France) and 2032 coin cell were used to carry out the electrochemical characterization. The two-electrode cells were assembled by sandwiching copolymers between the working electrode and a lithium metal foil, which served as the reference and the counter electrode, simultaneously. Moisture contaminations were avoided by assembling the cells inside the Ar-filled glove-box. Stainless steel and copper disks were used as working electrodes during anodic and cathodic stability measurements, respectively. To evaluate anodic limits, potential sweeps were carried out between OCV and 5.5 V vs. Li^+/Li at a constant rate of 0.2 mV s^{-1} . To determine cathodic limits, potential sweeps were performed between OCV and -0.5 V vs. Li^+/Li at the same constant rate.

The lithium-ion transference number (t_{Li^+}) was determined at 70 °C in the $\text{Li}/\text{coPIL}/\text{Li}$ cell with total applied potential bias of 0.16 V (ΔV) in accordance to the method described by Evans *et al.*⁵⁶ and using the following equation:

$$t_{\text{Li}^+} = \frac{I^s(\Delta V - I^o R_1^o)}{I^o(\Delta V - I^s R_1^s)} \quad (1)$$

, where t_{Li^+} is the Li transference number, ΔV is the potential applied across the cell, R_1^o and R_1^s are the initial and steady-state resistances of the passivating layer, I^o and I^s are the initial and steady-state currents.

RESULTS AND DISCUSSION

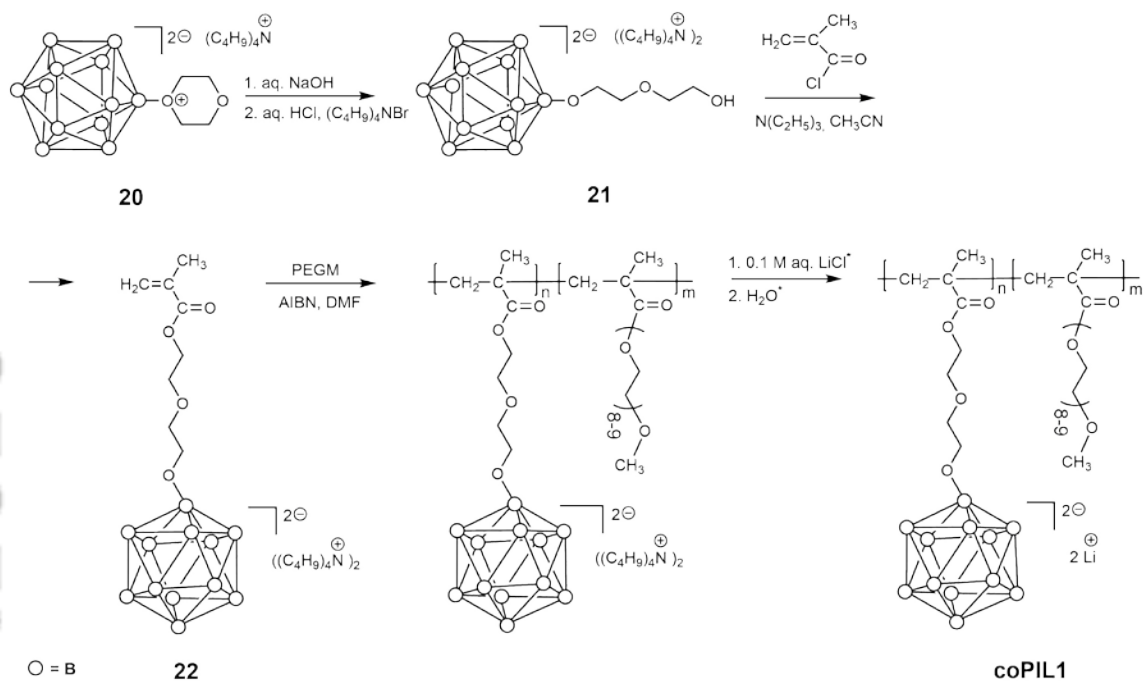
Synthesis of ionic monomers

Designing novel anionic monomers (Scheme 4, **22** and Scheme 5, **24**) the previously established relationships between the structure of PILs and their ionic conductivity were taken into account. It is prominent that for higher bulk ionic conductivity of polyelectrolyte the chemically attached anion should be placed at the end of the comb-like side chain, while the later should be long enough to enable sufficient ion motion and at the same time it should contain polar ethylene

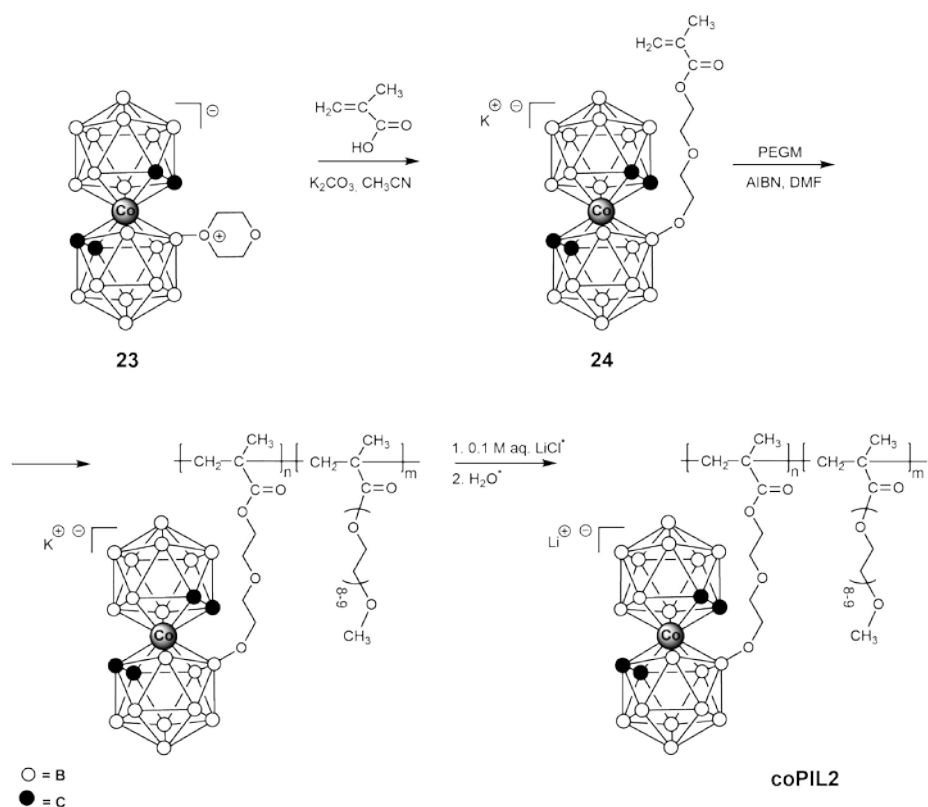
oxide fragments to increase dissociation of ions^{8,10,11,13,16,57,58}. Keeping this in mind, two main points were applied for the development of ionic monomers **22** and **24** in the present work: the insertion of flexible $-(O-CH_2CH_2)_2-O-$ spacer between the boron anion and methacrylic reactive group in contrast to previously reported polyelectrolytes (Scheme 1, **5**, **7**, **9-11**) and the introduction of cheap polyhedral boron anions instead of expensive fluorinated TFSI anion (Scheme 1, **7-11**). The synthetic strategy for the preparation of **22** and **24** included synthesis of dioxane-derivatives of *closo*-dodecaborate and cobalt bis(1,2-dicarbollide) anions with subsequent ring opening via nucleophilic attack on the positively charged oxygen atom (Schemes 4 and 5). In case of monomer **22** the attack was performed by NaOH and the intermediate product **21** was isolated from aqueous medium by ion exchange with hydrophobic tetrabutylammonium cation (Scheme 4). The last step consisted in acylation of **21** with methacryloyl chloride in the presence of trimethylamine as proton acceptor. After washing with water and extraction with methylene chloride monomer **22** was isolated in 98% yield.

For monomer **24** the reaction pathway was simplified (Scheme 5). As bis(1,2-dicarbollide) anion is only mono-charged and hydrophobic enough to be extracted from aqueous solution into the organic solvent, both the nucleophilic attack by K_2CO_3 and the esterification reaction with methacrylic acid were performed simultaneously to produce **24** in quantitative yield and high purity.

Monomer **22** bearing two common for ILs tetrabutylammonium cations was obtained as white crystalline solid showing relatively low melting point of 70-74°C. In contrast, monomer **24** was prepared in potassium salt form that insured its solid nature and high melting point (176-178°C). The structure and purity of monomers **22** – **24** were proved by 1H and ^{11}B NMR, IR spectroscopy and elemental analysis. FTIR spectra of monomers showed the characteristic absorption bands at ~2960, 2900-2880 (CH_2 stretching), 2470 and 2570 (B-H vibrations), 1715 (C=O from ester), 1630 (C=C vibrations) and at 1150-1050 cm^{-1} (set of bands identified as vibrations of the C-O-C bond in oxyethylene units). Monomer **24** differed by the presence of a band at 3030 cm^{-1} that was attributed to the vibration of the CH groups in carborane fragment⁵⁵.



Scheme 4. Synthetic pathway for the preparation of novel ionic-liquid-like monomer **22** and polymer **coPIL1** ($n:m = 1:7.2$ by mol, *dialysis).



Scheme 5. Synthetic pathway for the preparation of novel ionic-liquid-like monomer **24** and polymer **coPIL2** (n:m = 1:7.2 by mol, *dialysis).

Free radical copolymerization

It was formerly demonstrated by our group that the highest ionic conductivity for methacrylate SICs can be achieved by copolymerization of Li-ion containing monomers with poly(ethylene glycol) methyl ether methacrylate (PEGM) in [ILM]:[PEGM] molar ratio equal to 1:7.2¹⁹. PEGM was chosen due to the presence of oxyethylene fragments in its side chain that, by analogy to PEO⁵⁹, should improve the solubility of ionic species, facilitate their dissociation, and promote an increase in the conductivity of the resultant copolymers. Moreover, copolymerization with PEGM often allows for the decrease in glass transition temperature of resultant polyelectrolytes, that in its turn affects their ionic conductivity increasing it to a high level^{17,19,19,60}. In view of this and with the aim of appropriate comparison with previously published methacrylate SICs, monomers **22** and **24** were copolymerized with PEGM using the optimal ratio of 1:7.2 (Table 1, entries 1 and 4). Free radical copolymerization was carried out in DMF with 1 wt.% of AIBN at 60°C. The applied concentration was kept at 33 wt. % ([monomers]:[DMF]=1:2 by wt.) leading to the preparation of polymers in 60-75% yield. Although the increase in concentration to 50 wt.% allowed the enhancement of polymer isolated yields up to 85%, it mostly resulted in the obtaining of insoluble (cross-linked) polyelectrolytes. The isolated copolymers further undergone dialysis against 0.5M LiCl aqueous solution to gain polyelectrolytes in a lithium form (Schemes 4 and 5). Finally, the resultant copolymers were isolated by freeze-drying and were abbreviated as **coPIL1** and **coPIL2**, respectively.

The structure and purity of **coPIL1** and **coPIL2** were supported by elemental analysis, ¹¹B, ⁷Li NMR and IR spectroscopy. FTIR spectra of copolymers showed the absorption bands at 2870 cm⁻¹ assigned to CH₂ stretching, 2470 and 2560 cm⁻¹ attributed to B-H vibrations, 1730 cm⁻¹ recognized as vibrations of the C=O group and at 1100-1040 cm⁻¹ assigned to the C-O-C vibrations of the ether groups. The absence of residual monomers was proven by the disappearance of the band at 1636 cm⁻¹ (C=C). The ion exchange from ammonium or potassium to lithium cations in **coPIL1** and **coPIL2**, respectively, was supported by the presence of a single peak in ⁷Li NMR. The coincidence of both the position and the number of signals in ¹¹B NMR for **coPIL1** and **coPIL2** stated for the preservation of borate anions after polymerization and subsequent treatment of copolymers. The results of the elemental analysis allowed to determine the copolymers composition, which was found to be [ILM:PEGM] = 1:8.5 and 1:6.8 for **coPIL1** and **coPIL2**, correspondingly.

CoPIL1 and **coPIL2** were found to have similar solubility in organic solvents: both of them were readily soluble in water, acetone, acetonitrile, and aprotic polar solvents (DMF, DMSO, DMAc). They were found to be insoluble in aliphatic hydrocarbons (pentane, hexane) and diethyl ether. The measured coPILs solution inherent viscosities varied from 0.62 to 0.90 dL g⁻¹ indicating relatively high molecular weights (Table 1, entries 1 and 4), that was further proved by investigation of their molar masses via GPC. The study of **coPIL1** and **coPIL2** by GPC in aq. 0.1M LiCl solution provided M_w values as high as 3.16×10^5 and 2.55×10^5 g mol⁻¹, respectively.

Thermal properties of copolymers

Thermal properties of **coPIL1** and **coPIL2** were studied by DSC and TGA (Table 1 and Fig. 1). They were estimated in comparison with copolymer **7** as a “classical” representative of TFSI based SICs (Table 1, entry 5). With the aim of correct comparison copolymer **7** was prepared using the same [ILM]:[PEGM] = 1:7.2 molar ratio. The observed T_g values for copolymers were found to be in a similar range and can be arranged in the following order:

$$T_{g\ 7} (-57) \leq T_{g\ coPIL1} (-53) \leq T_{g\ coPIL2} (-49^\circ\text{C})$$

As the values of the T_g were very close it can be concluded that the heat resistance of the copolymers is independent of the nature of bonded anion and is mainly driven by the presence of PEGM units (Table 1, entries 1, 4-6).

The thermal degradation behavior of coPILs was assessed by TGA (Table 1 and Fig. 1). The weight loss profile of **coPIL1** and **coPIL2** revealed one-step degradation mechanism. It was found that the weight loss onset temperature T_{onset} is strongly dependent on the nature of the chemically bonded anion and can be summarized as follows:

$$T_{onset\ coPIL1} (210) > T_{onset\ 7} (180) > T_{onset\ coPIL2} (155\ ^\circ\text{C})$$

Thus, the closo-dodecaborate anion imparted highest thermal stability, while cobalt bis(1,2-dicarbollide) anion showed the lowest one. It is worth mentioning, that organic salts bearing TFSI or BF₄ anions usually demonstrated highest thermal stability among ILs⁶¹. The same principle was found to be correct in the case of PILs as well^{62,63}. However, in this work, for the first time the thermal stability of TFSI-based copolymer **7** was overcome by **coPIL1** bearing closo-dodecaborate anion and having the same composition. Such result represents particular interest for application in lithium batteries as the thermal stability of conventional liquid electrolytes based on organic solvents usually does not exceed 140°C under inert atmosphere⁴.

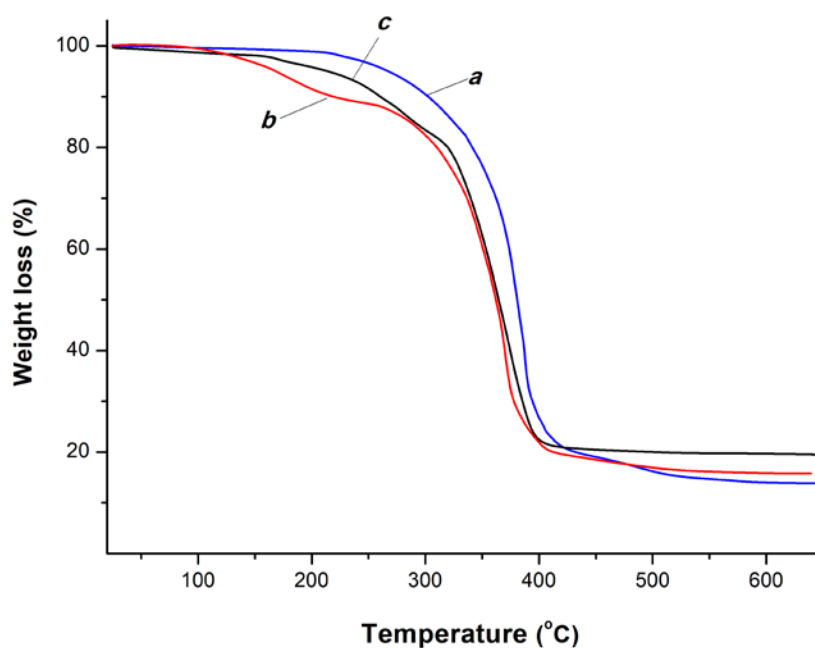


Figure 1.

Electrochemical properties of copolymers

Ionic conductivities of the single-ion conducting copolymers were measured by electrochemical impedance spectroscopy (EIS) in the temperature range between 20 and 100°C (Fig. 2). Conductivity values determined at 25°C ranged from 6.3×10^{-8} to 9.1×10^{-7} S cm⁻¹ in the following order (Table 1):

$$\sigma_{\text{DC}}^{\text{coPIL2}} > \sigma_{\text{DC}}^{\text{coPIL1}} > \sigma_{\text{DC}}^{\text{7}}$$

Both newly synthesized coPILs demonstrated ionic conductivity higher than that reported previously for TFSI-based copolymer **7** (Table 1, entries 1, 4 and 5). Probably, such result can be explained by higher delocalization of the negative charge in borate anions⁶⁴ and as a result by high counter-cations mobility. The conductivity of the monocharged **coPIL2** was found to be nearly one order of magnitude higher at 25°C than that of **coPIL1** bearing two lithium cations (Table 1, entries 1 and 4). Similarly, an enhanced mobility has previously been reported in the monovalent case vs. the divalent case for the borate metal salts ($\sigma_{\text{MCB}_9\text{H}_{10}} \gg \sigma_{\text{M}_2\text{B}_{12}\text{H}_{12}}$, M = Li or Na⁶⁵). This may be understood in terms of the charge densities associated with these objects. For a complex anion of a given size, the charge density will increase as a function of valency, resulting in a stronger binding of the Li⁺ and disfavoring their mobility. From the standpoint of Li⁺ mobility, therefore, it

may be argued that it is better to distribute a single charge over a larger structure than multiple charges over smaller structure.

The conductivity of **coPIL1** and **coPIL2** as a function of inverse temperature between 20 and 100 °C is shown on Figure 2. For both copolymers the conductivity increased with the rise in temperature and reached the level of 10^{-5} S cm $^{-1}$ at 98 and 63°C for **coPIL1** and **coPIL2**, correspondingly. However, conductivity vs temperature dependence did not follow linear Arrhenius behavior. This in its turn means that for **coPIL1** and **coPIL2** the lithium ion diffusion results from both mechanisms: hopping of lithium cations between borate anions and from local segmental motion of the side chains bearing coordinative oxyethylene fragments.

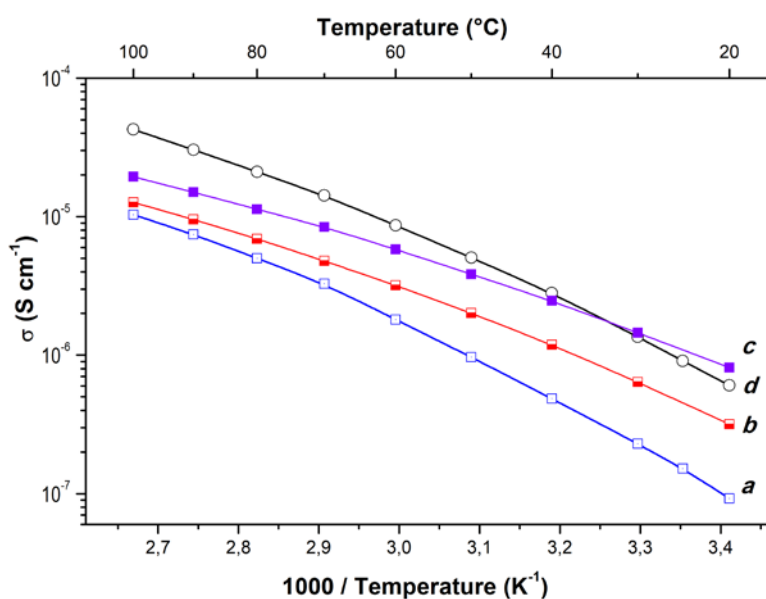


Figure 2.

The lithium-ion transference numbers for copolymers were evaluated at 70 °C using the method of Evans et al.⁵⁶ The test was performed by measuring of the impedance of symmetrical Li/coPIL/Li cell before and after polarization experiment (Table 1).

$$t_{Li^+} \text{ 7 (0.83) } > t_{Li^+} \text{ coPIL2 (0.61) } > t_{Li^+} \text{ coPIL1 (0.39)}$$

The transference number value was found to be 0.39 and 0.61 for **coPIL1** and **coPIL2**, respectively. For both copolymers it was lower than for TFSI-based copolymer **7**. This can be explained by several reasons. Firstly, as it was discussed in conductivity section from the standpoint of Li $^{+}$ mobility it will be preferable to have a single charged anion over the multicharged one. Secondly, the borate anions are attached to the main polymer chain by a flexible oxyethylene

spacer that allows them to move or in the other words it grants the short-range motion of negative charges. And if compared the spacer in **coPIL1** and **coPIL2** is more flexible than in copolymer **7**, thus leading to higher local motions of the borate anions. Finally, due to the fact that tests were conducted well above the glass transition temperature of copolymers the additional motions of negative charges may arise from the segmental motion of the polymer backbone.

The electrochemical stability of the copolymers was studied by cyclic voltammetry at 70°C (Fig. 3). Figures **3a** and **3c** demonstrate the cathodic scans for **coPIL1** and **coPIL2** against the copper electrode. For both copolymers two reversible redox peaks at -0.40 and between 0.20 - 0.35 V versus Li^+/Li were observed that are associated to the reversible lithium plating and stripping at the copper electrode surface. For **coPIL2** an additional irreversible peak was found at 1.5 V, which can be attributed to the oxidation processes of carbon atoms in the anion. During continued scanning, the intensity of this anodic peak significantly reduces. The oxidation potentials for **coPIL1** and **coPIL2** against stainless steel electrode were found to be around 4.0 V vs. Li^+/Li (Fig. 3a and 3c) and the overall evolution of electrochemical stability of coPILs can be summarized as follows:

$$\text{ESW } \mathbf{7} (4.3) \geq \text{ESW } \mathbf{coPIL2} (4.1) \geq \text{ESW } \mathbf{coPIL1} (4.0)$$

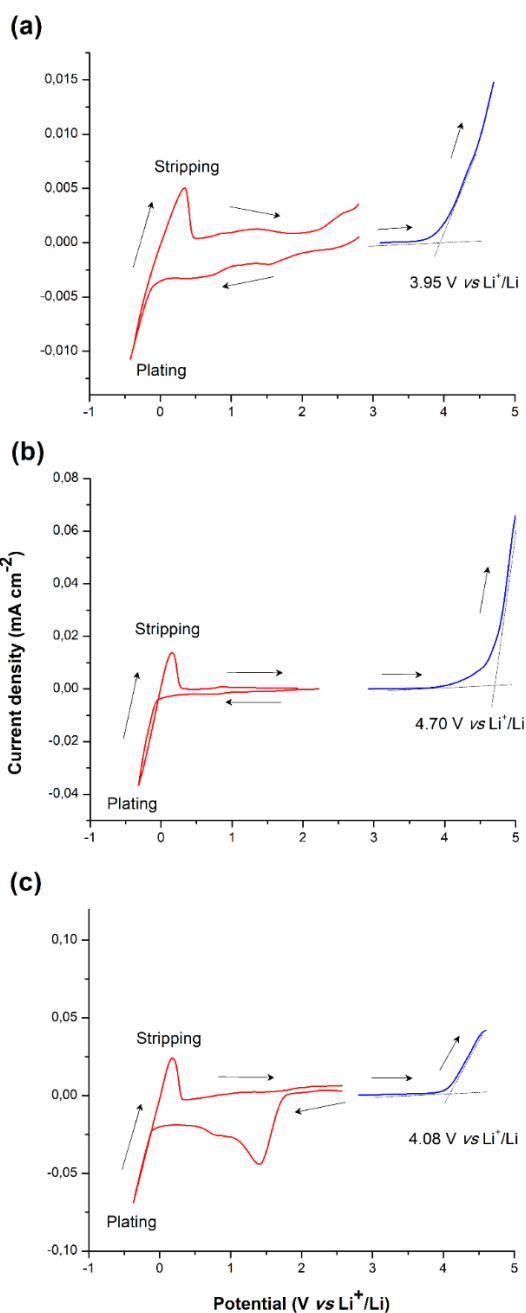


Figure 3.

Properties of the filled copolymers

Although copolymers synthesized in this work demonstrated conductivity of $\sim 10^{-6} \text{ S cm}^{-1}$ at 25°C such result is still not sufficient for their industrial application. To increase its ionic conductivity the **coPIL1** was filled with plasticizer, namely propylene carbonate (PC). Two

concentrations were studied: 20 and 40 wt% (Table 1, entries 2-3). Both filled polyelectrolytes were found to be thermally stable up to 100°C in air (Fig. 1, curve 2). TGA curves of the samples showed two-step degradation with the first step attributed to the loss of PC (at 100 °C) and the second step at around 245°C assigned to the degradation of the polymeric matrix. The incorporation of 20 wt% of PC in **coPIL1** resulted in significant decrease in its T_g down to -65°C and in nearly half order of magnitude increase in conductivity. Furthermore, the presence of 40 wt% of PC allowed **coPIL1** to reach the level of 10^{-6} S cm⁻¹ at 25°C, without changing the shape of the temperature vs conductivity curve (Fig. 2, curve 3). Finally, the addition of even 20 wt% of PC to **coPIL1** significantly improved its anodic limit up to 4.7 V vs. Li⁺/Li (Fig. 3b).

CONCLUSIONS

To expand the chemistry of single-ion conducting poly(ionic liquid)s, the synthetic route for the preparation of two novel ionic liquid like monomers bearing borate anions was suggested. The monomers differed by the nature of chemically bonded anion (*closo*-dodecaborate or cobalt bis(1,2-dicarbollide)) and by valency (divalent-anion and monovalent one). Their simple free radical copolymerization with poly(ethylene glycol) methyl ether methacrylate (PEGM) in a previously determined optimal ratio ([ILM]:[PEGM] = 1:7.2 by mol) and subsequent ion exchange provided new Li-ion conducting polyelectrolytes. As previously observed for other families of single-ion conductors, their properties such as solubility, viscosity, thermal stability, T_g , ionic conductivity and electrochemical stability are affected by the constitution of ILMs and the nature of the attached anion. The monovalency of the anion and the resulting 1:1 cation:anion stoichiometric ratio were found to be preferable for enhancing of Li⁺ mobility and increasing of overall ionic conductivity.

Among the advantages of the novel SICs are: a) low cost and availability in comparison with SI-based polyelectrolytes; b) low T_g values of around -50°C; c) conductivity level of $\sim 10^{-6}$ S cm⁻¹ at 25°C; d) sufficient thermal stability (T_{onset} = 155-210°C) and e) good electrochemical stability window versus Li⁺/Li (4.0-4.1 V). Their performance can be remarkably improved by the incorporation of propylene carbonate (PC), thus making them potential candidates for the next generation of safe, cost-effective and environmentally friendly lithium polymer batteries.

ACKNOWLEDGEMENTS

Merlin Cotessat is grateful to the Luxembourg Institute of Science and Technology (LIST) for the sponsorship of his internship and work in the Materials and Research Technology Department

(MRT). This work was in part supported by a grant from Russian Foundation for Basic Research RFBR (project no. 18-33-20108_mol_a_ved). Authors would like to thank Dr. Dmitry S. Perekalin (A.N. Nesmeyanov Institute of Organoelement Compounds Russian academy of sciences (INEOS RAS)) for the fruitful discussions on boron chemistry. Authors would also like to express their special gratitude to Dr. Elie-Elisée Paillard (Helmholtz-Institute Münster, Forschungszentrum Jülich GmbH) for the kind supply of high-quality lithium foil.

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Table 1. Selected coPIL's properties

N°	Polymer	Additive (wt%)		η_{inh} (dL g ⁻¹) ^a	$M_w \times 10^{-5}$ (g mol ⁻¹) ^b	PDI ^b	T_g (°C) ^c	T_{onset} (°C) ^d	σ at 25 °C (S cm ⁻¹)	Electrochemical window, (V) ^e	t_{Li+} ^f
1	coPIL1	-	-	0.90	3.16	3.54	-53	210	1.5×10^{-7}	4.0	0.61
2	coPIL1	PC	20	-	-	-	-65	100	4.5×10^{-7}	4.7	-
3	coPIL1	PC	40	-	-	-	-73	100	1.1×10^{-6}	-	-
4	coPIL2	-	-	0.62	2.55	4.19	-49	155	9.1×10^{-7}	4.1	0.39
5 ^g	7¹⁹	-	-	-	4.57	4.35	-57	180	6.3×10^{-8}	4.3	0.83
6 ^g	polyPE GM	-	-	-	0.94	2.22	-62	160	-	-	-

^a For the solution of 0.05 g of coPIL in 10.0 mL of aq. 0.5M LiCl at 25.0°C.

^b By GPC in 0.1 M LiCl solution of water/ACN (4:1 v/v) at 25°C with PEG standards calibration.

^c By DSC.

^d By TGA.

^e Determined by cyclic voltammetry at 70°C (stainless steel as the working electrode and Li foil as the counter and reference electrodes, scan rate 0.2 mV s⁻¹).

^f Determined at 70°C using the method of Evans *et al.* ⁵⁶.

^g For comparison form reference¹⁹.

Figure 1. TGA traces for **coPIL1** (a), **coPIL1** filled with 20 (b) wt.% of propylene carbonate and for **coPIL2** (c).

Figure 2. Ionic conductivity vs temperature dependence for **coPIL1** (a), **coPIL1** filled with 20 (b) and 40 (c) wt.% of propylene carbonate and for **coPIL2** (d).

Figure 3. Electrochemical stability windows for **coPIL1** (a), **coPIL1** filled with 20wt% of PC (b) and **coPIL2** (c) at 70 °C (stainless steel as the working electrode and Li foil as the counter and reference electrodes, scan rate 0.2 mV s⁻¹).